Results: Language, Topics, and Embeddings

RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.

Table 3. NMF topics extracted from the corpus.

RQ3: Topical and Linguistic Structure I

Topic	Top terms
0	modafinil, mg, kg, effects, rats, mice, induced, sleep
1	fatigue, patients, placebo, modafinil, scale, armodafinil, treatment, depression
2	h4, methods, results, conclusion, 95, analysis, ci, risk
3	use, cognitive, drugs, adhd, studies, drug, cocaine, methylphenidate
4	sleep, narcolepsy, sleepiness, cataplexy, daytime, patients, eds, excessive

RQ3: Topical and Linguistic Structure I

The topics reveal the thematic structure of the literature: Topic 0 reflects preclinical dose-response pharmacology in animal models (mg/kg dosing, dopamine activity); Topic 1 captures clinical fatigue and mood trials (placebo-controlled patient studies, armodafinil); Topic 2 collects structured-abstract and meta-analytic language (methods, results, 95% confidence intervals, risk estimates); Topic 3 centres on cognitive enhancement and psychostimulant pharmacology (ADHD, methylphenidate, cocaine comparisons); and Topic 4 addresses sleep disorders (narcolepsy, excessive daytime sleepiness, cataplexy, hypersomnia). These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

RQ3: Topical and Linguistic Structure I

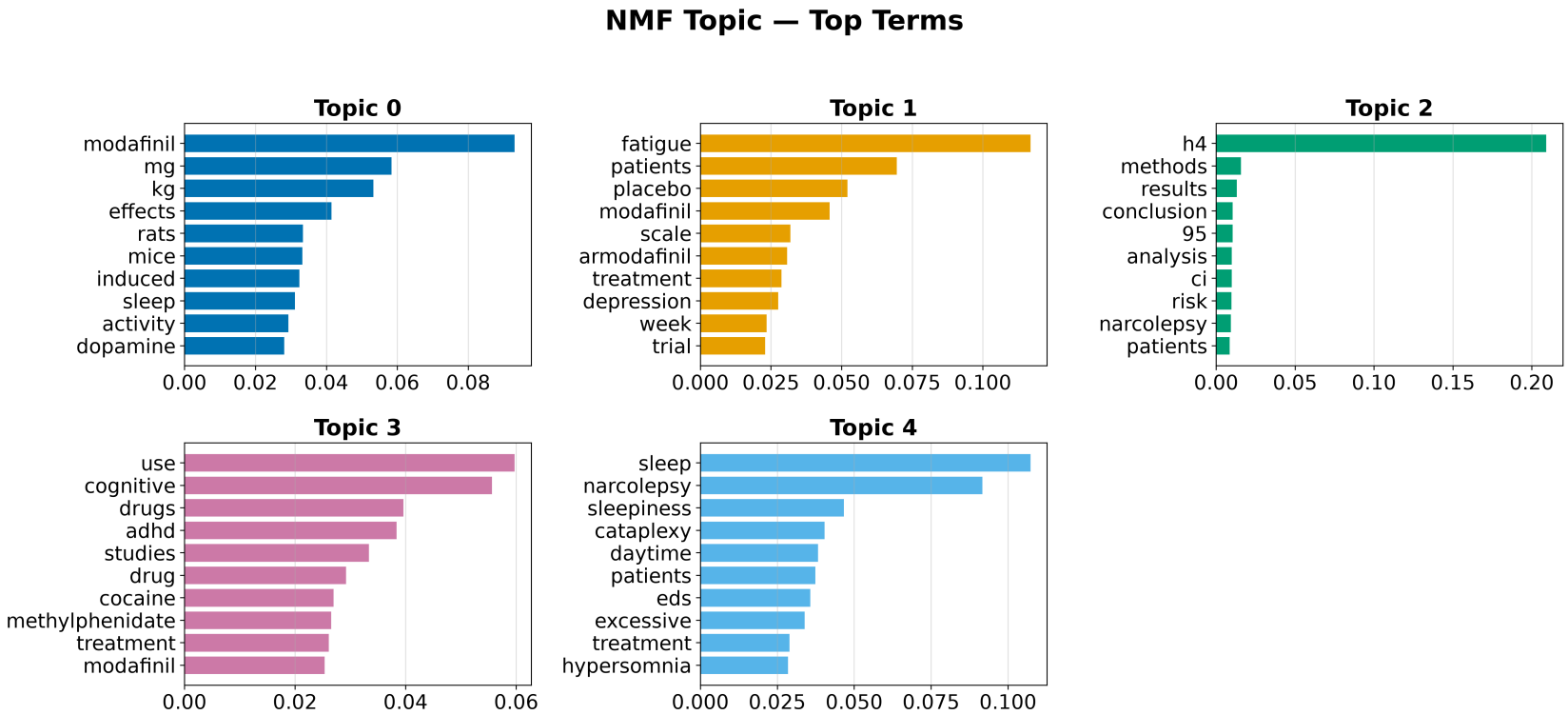


Figure 9: Topic-term bar charts for Modafinil. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

Document Embeddings I

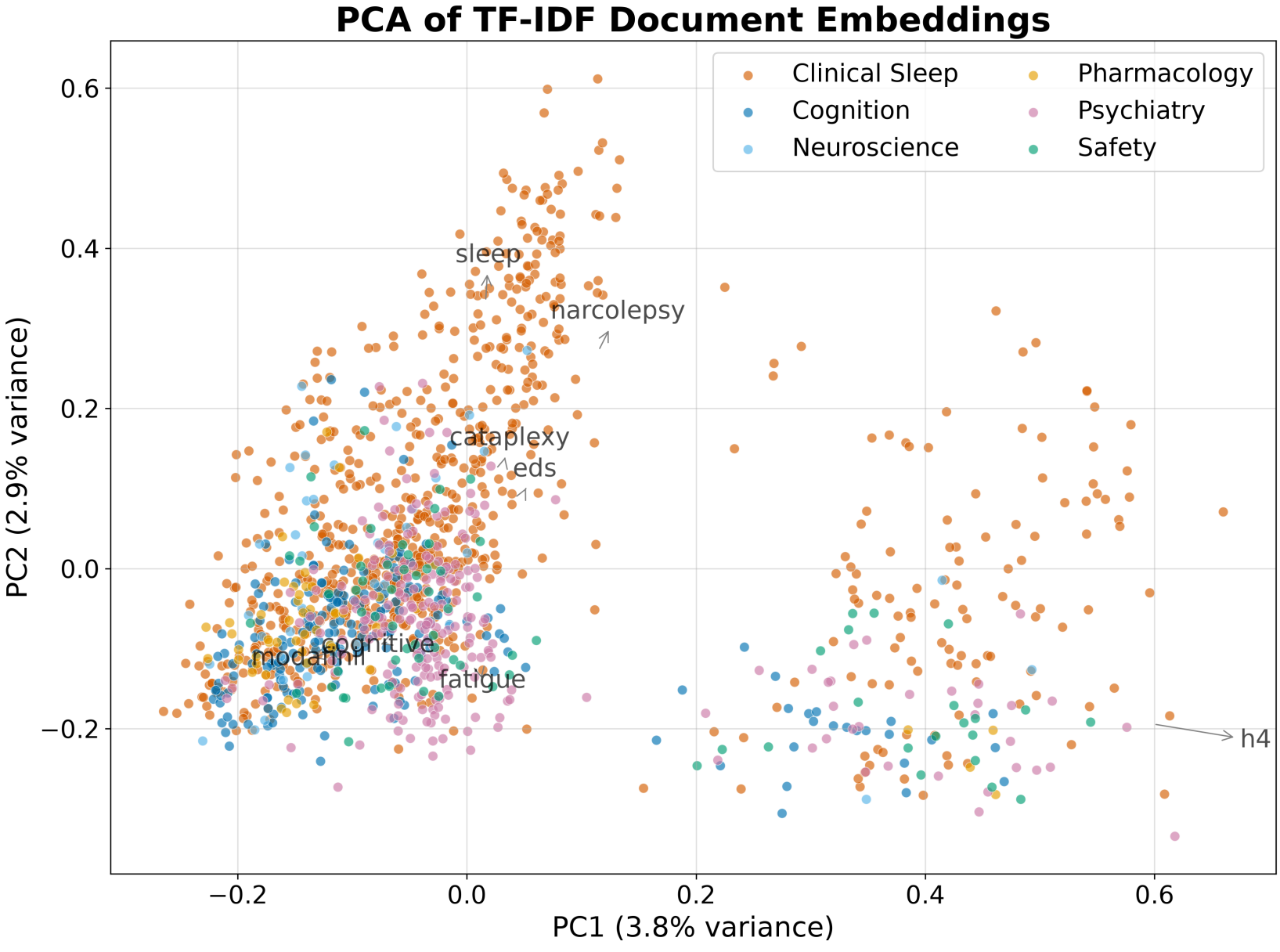


Figure 10: PCA projection of document embeddings for Modafinil. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

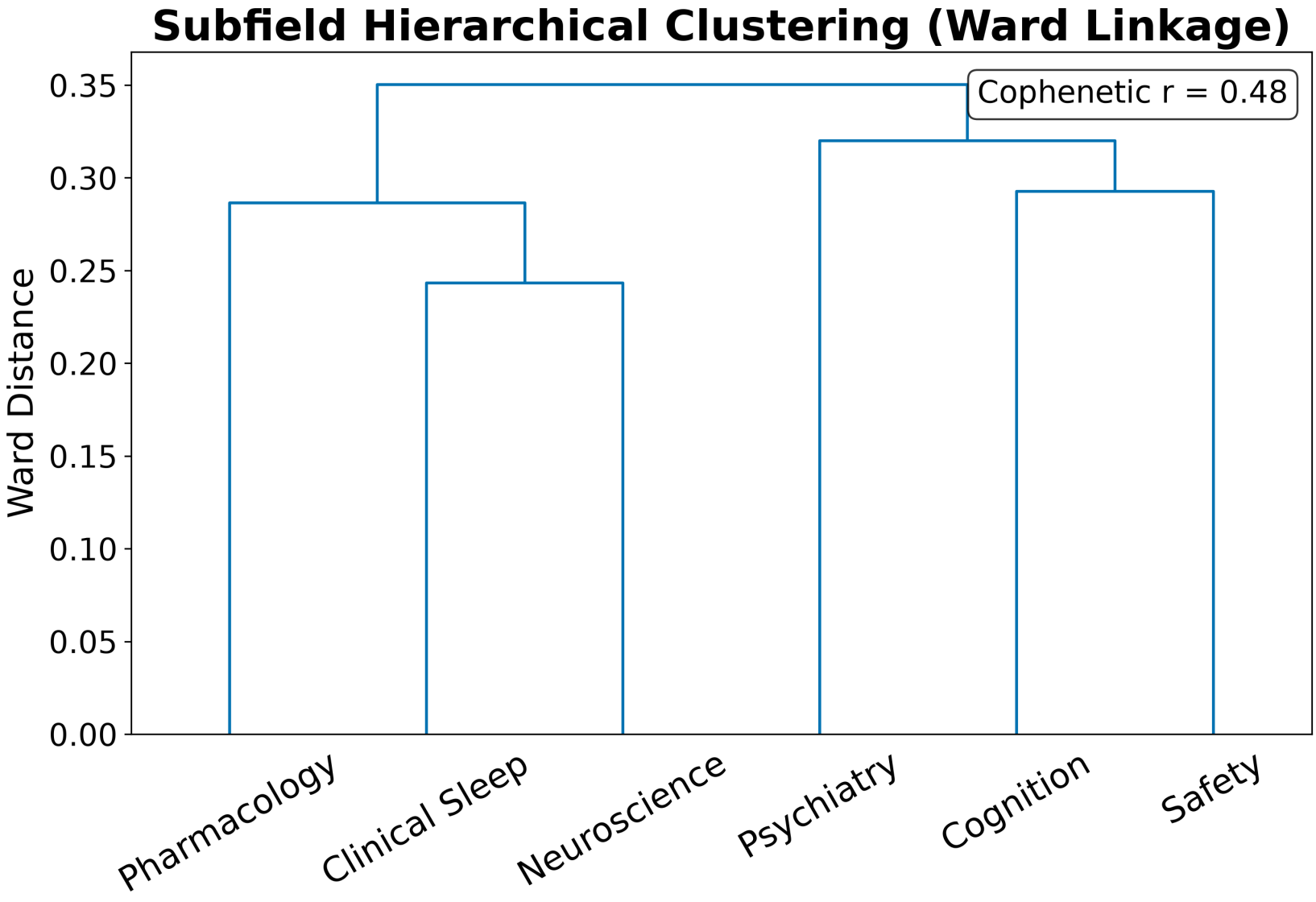

Subfield Hierarchical Clustering (Ward Linkage)
Cophenetic r = 0.48
Ward Distance
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Pharmacology
Clinical Sleep
Neuroscience
Psychiatry
Cognition
Safety

Figure 11: Hierarchical clustering dendrogram of subfield TF-IDF centroids. Leaves are the configured subfields; the tree shows the similarity structure between subfield centroid vectors.

Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

Term Analysis I

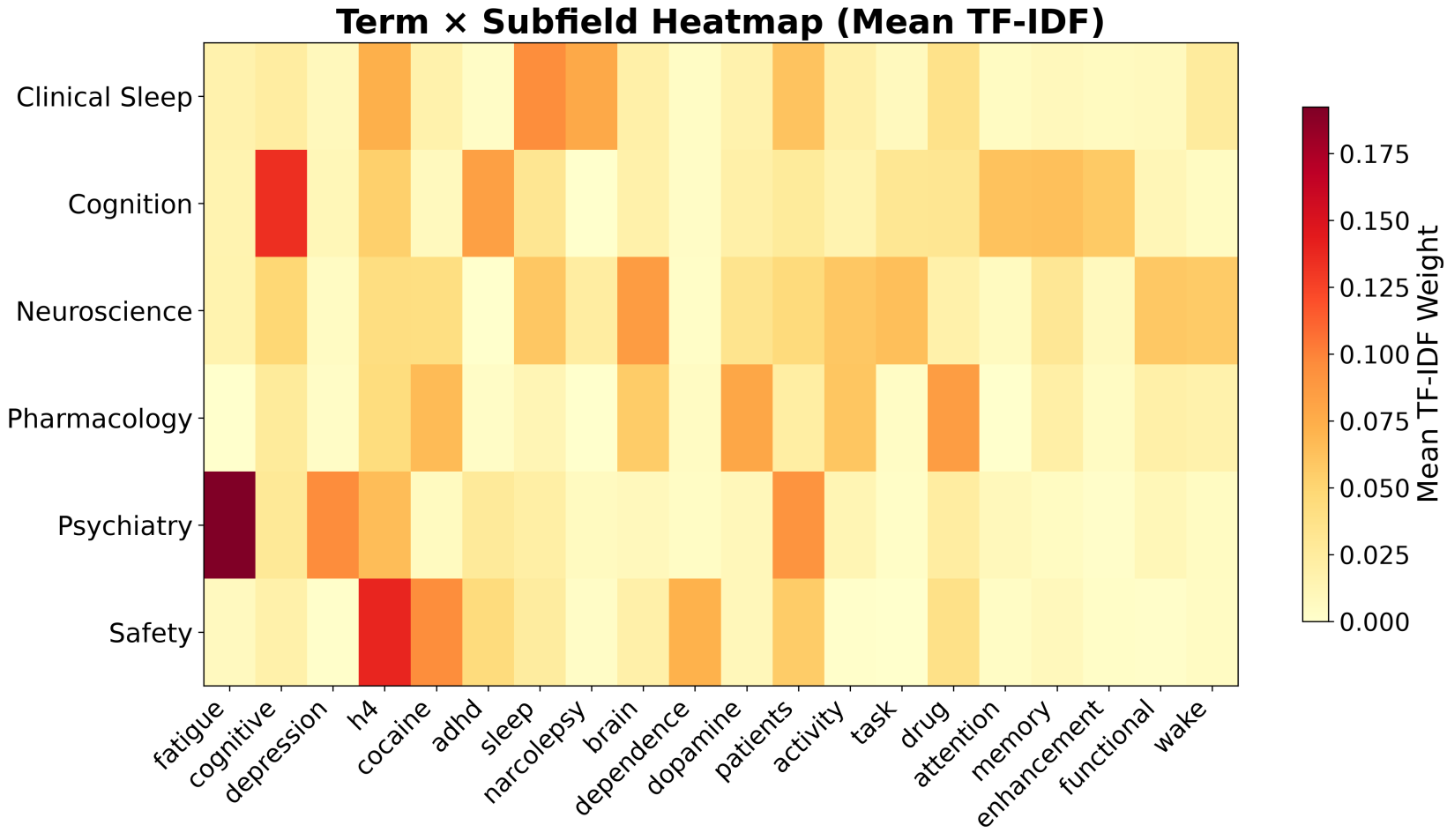


Figure 12: Term heatmap for Modafinil. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

Named entity extraction over the 1437 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

Table 4. Top named entities in abstracts.

Named Entity Analysis I

Entity	Frequency
ADHD	392
CI	343
EDS	326
OSA	272
MOD	181
ESS	148
RESULTS	148
MS	145

Named Entity Analysis I

Entity	Frequency
DAT	134
NT1	131
SD	130
CONCLUSIONS	129
MD	102
CE	101
IH	101

Named Entity Analysis I

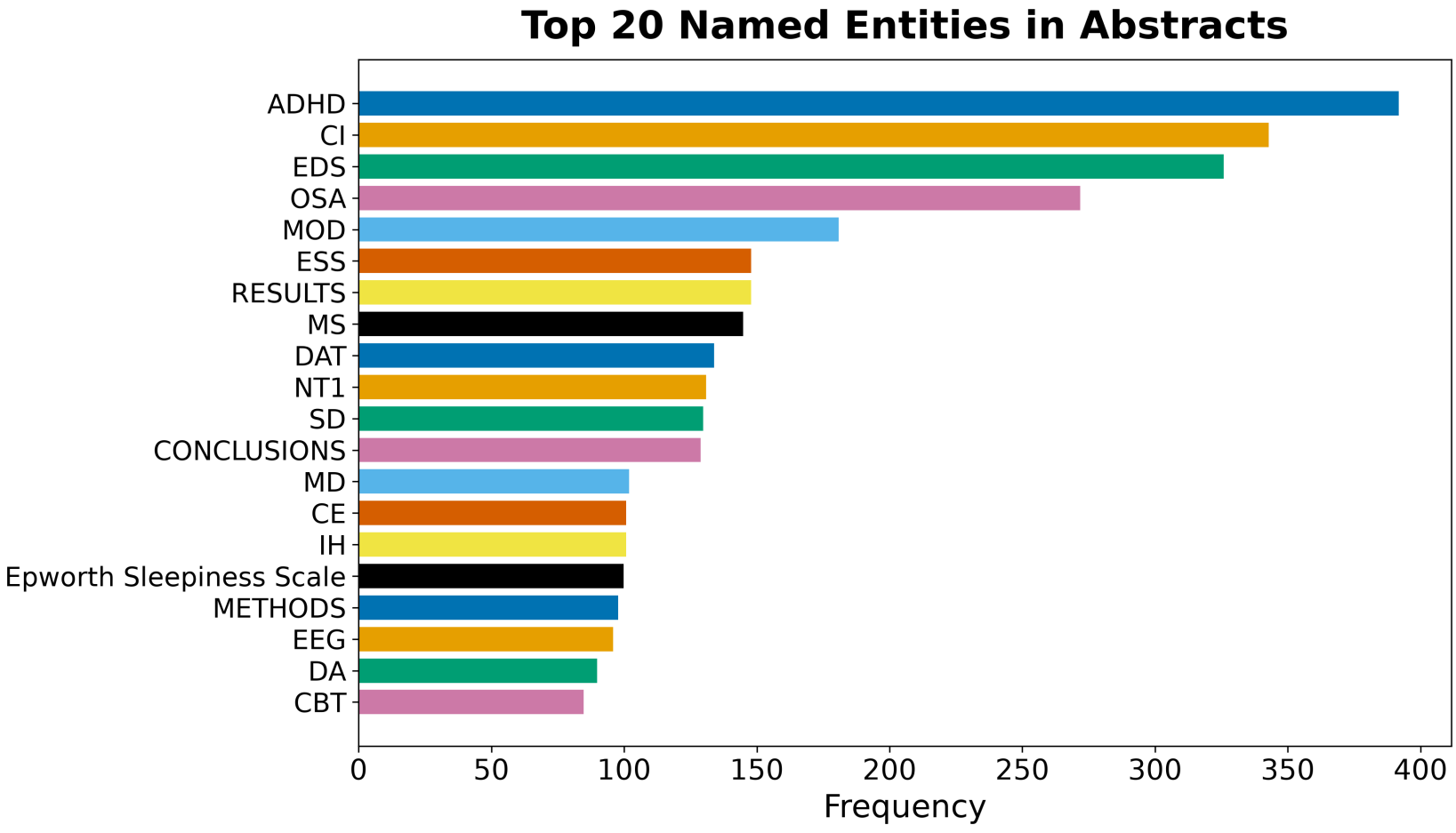


Figure 13: Top named entities for Modafinil. The horizontal bar chart shows the 20 most frequently extracted named entities from abstracts, revealing the dominant drugs, conditions, and concepts in the literature.

Table 5. Top keyphrases by TF-IDF score.

Named Entity Analysis I

Keyphrase	Score
available	0.3333
abstract	0.3333
abstract available	0.3333
content	0.1053
access	0.1053
sub	0.0881
md	0.0870
jama	0.0826

Named Entity Analysis I

Keyphrase	Score
cleveland	0.0763
modafinil	0.0741
depression	0.0741
cognitive	0.0714
substance	0.0694
drug	0.0667
conditions	0.0667

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Embedding Similarity and Clustering I

Paper A	Paper B	Similarity
doi:10.1176/appi.ajp.163.12.2184	doi:10.1176/ajp.2006.163.12.2184	1.0000
doi:10.1176/ajp.2006.163.12.2184	doi:10.1176/appi.ajp.163.12.2184	1.0000
doi:10.1345/aph.1h302	doi:10.1136/bcr.08.2011.4652	0.9660
doi:10.4088/jcp.09m05900gry	doi:10.1186/s40345-015-0034-0	0.9566
doi:10.1111/bdi.12859	doi:10.1111/acps.12712	0.9558
doi:10.1111/j.1360-0443.2008.02437.x	doi:10.1111/j.1465-3362.2012.00473.x	0.9536
doi:10.1513/annalsats.202006-696fr	doi:10.3760/cma.j.cn112147-20241118-00681	0.9532
doi:10.1164/ajrccm.164.9.2103032	doi:10.1093/sleep/28.4.464	0.9513

Embedding Similarity and Clustering I

Paper A	Paper B	Similarity
doi:10.1093/sleep/28.4.464	doi:10.1164/ajrccm.164.9.2103032	0.9513
doi:10.1002/cncr.25083	doi:10.1200/jco.2005.23.16_suppl.8012	0.9507

Embedding Similarity and Clustering I

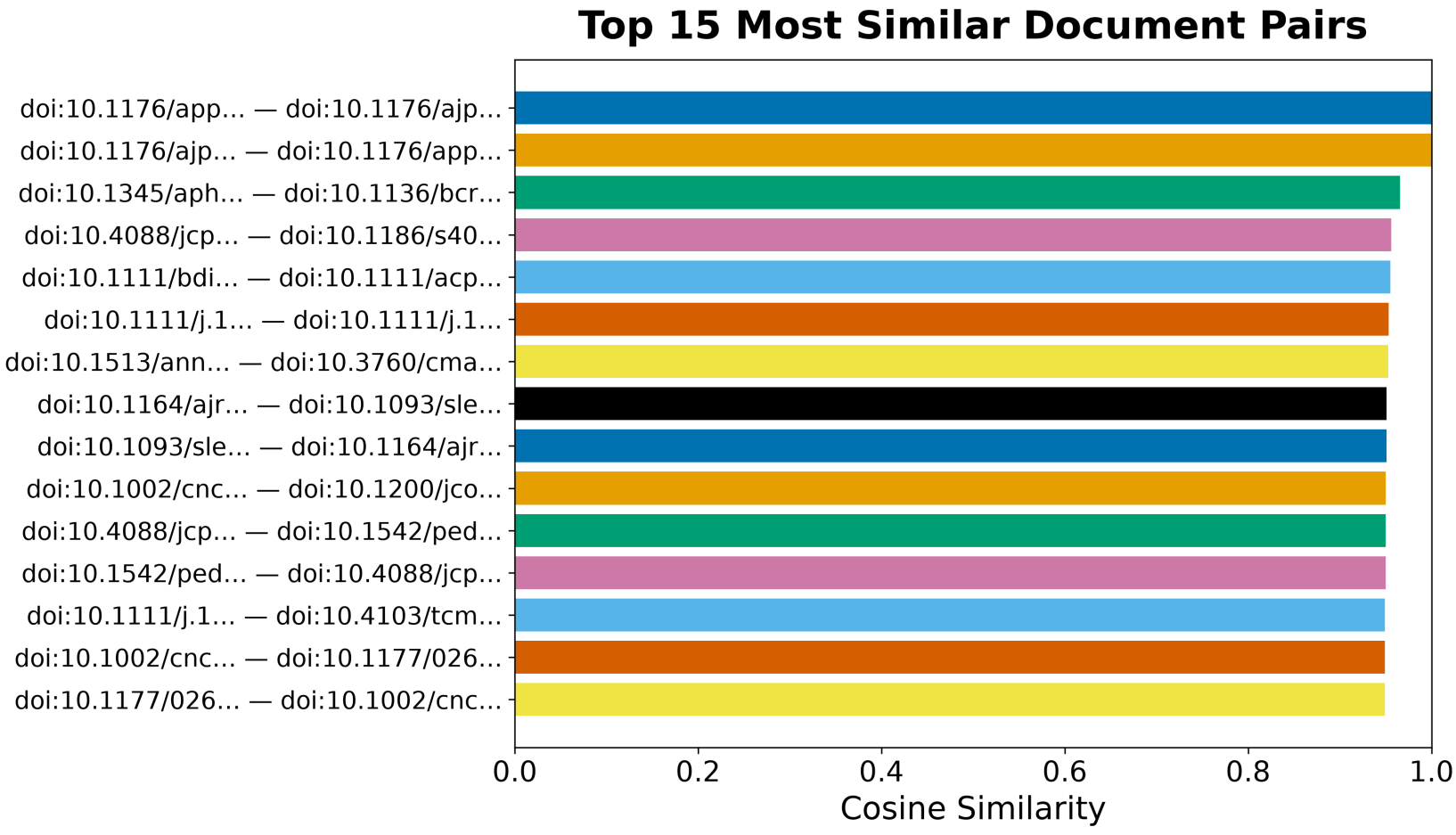


Figure 14: Document similarity for Modafinil. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

Embedding Similarity and Clustering I

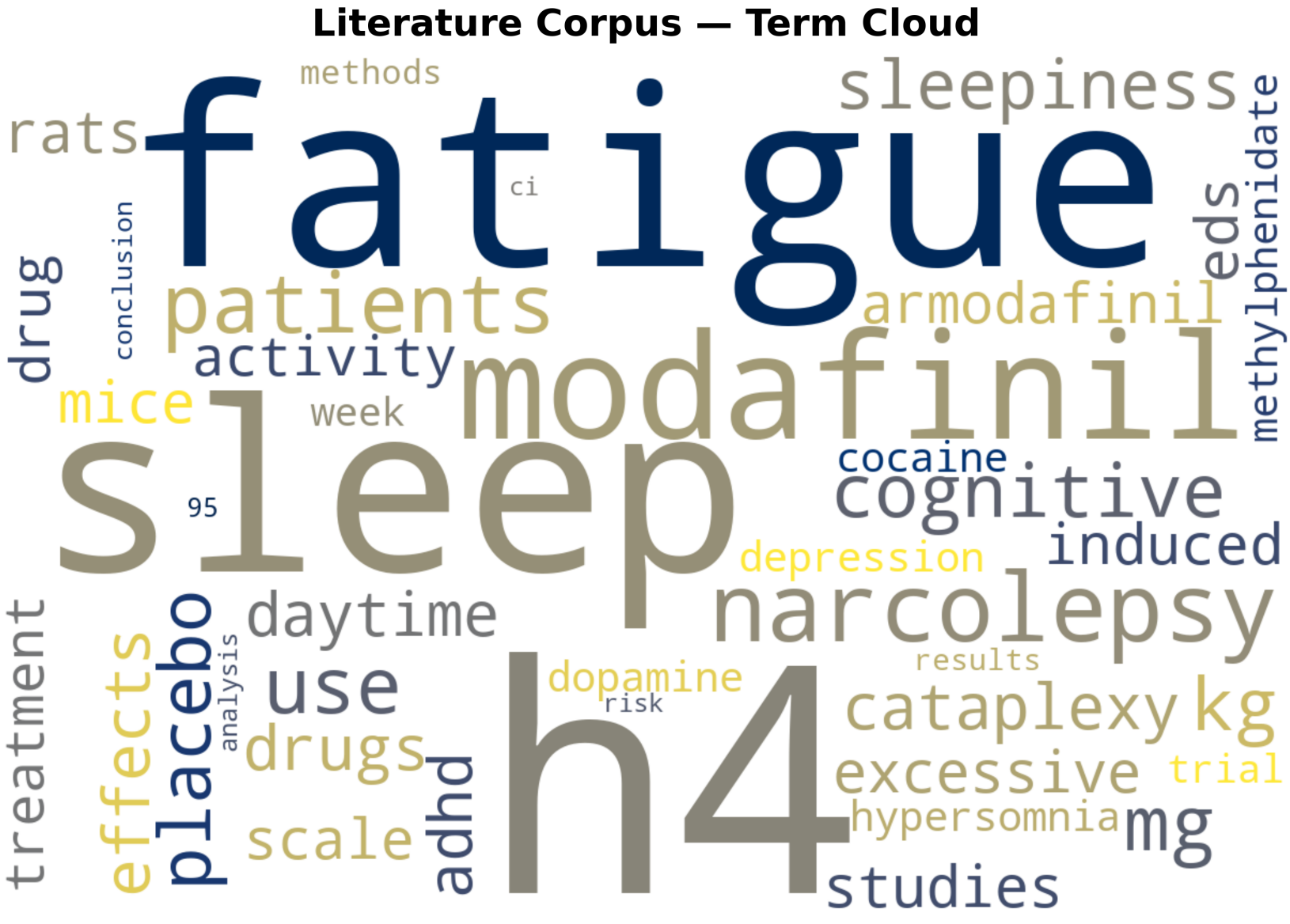


Figure 15: Term cloud for Modafinil. Term sizes are proportional to their topic-term weights from the discovered topics, providing a visual summary of the dominant vocabulary.

Embedding Similarity and Clustering I

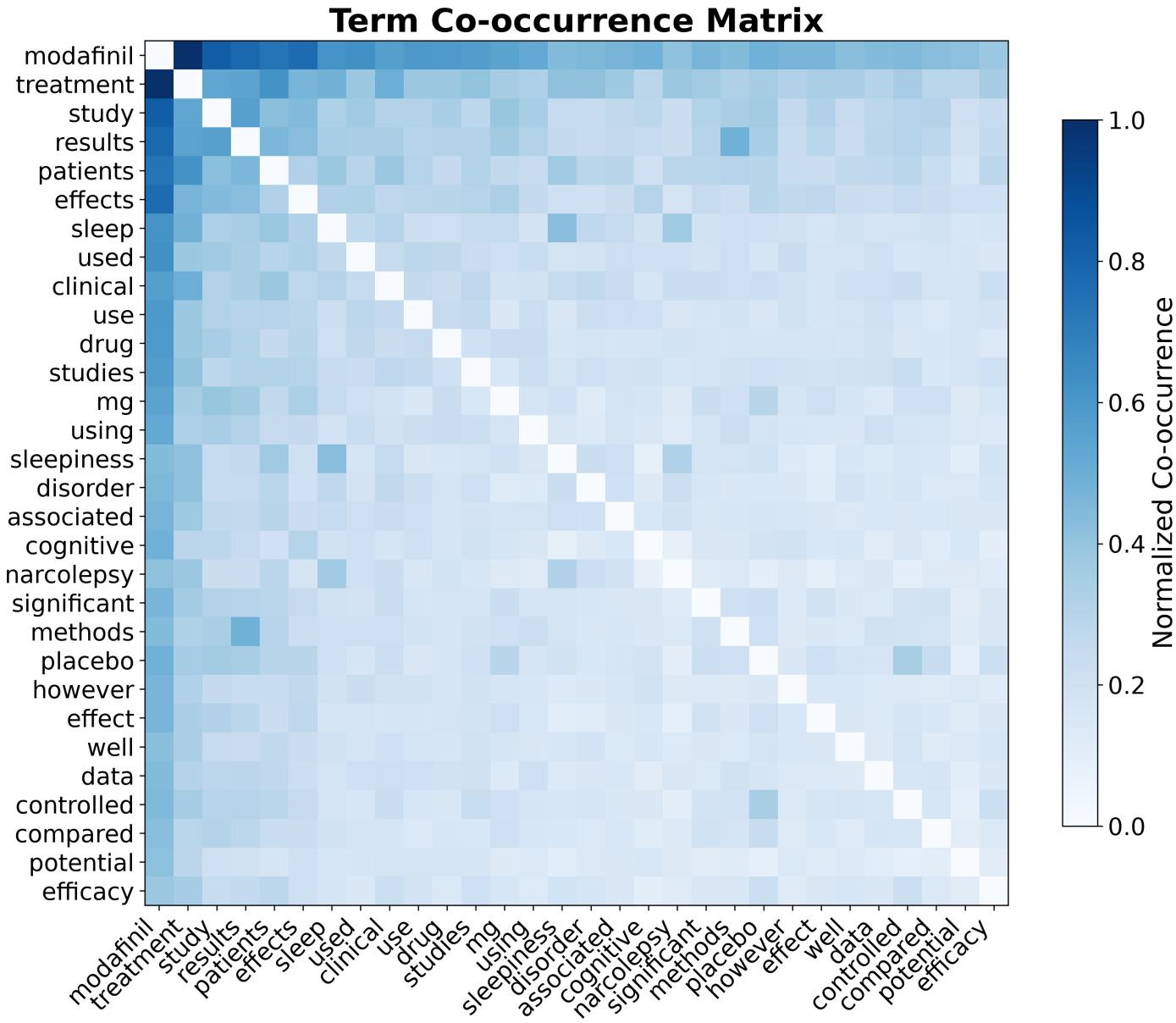


Figure 16: Term co-occurrence matrix for Modafinil. Each cell shows the normalized co-occurrence frequency of two terms within the same document, revealing which concepts tend to appear together in the literature.

Embedding Similarity and Clustering I

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.